TURBO: Top-Down Racing Game

Project documentation for the Afro-Asian Forum for Innovation and Technology (AFFIAT), Aswan Finals 2026. Built by Jomana and Ali, Team NINJA.PY.



> Every number in this document (levels, speeds, prices, signs, files) was checked against the code in this repository. If you change the game, update the document and run `python docs/build.py` to rebuild the PDF, the poster and the slides.

1. Project Overview | نظرة عامة على المشروع

Field	Value
Project Title	TURBO: Top-Down Racing Game
Project Type	Desktop game (Python + Pygame Zero)

Field	Value
Description	TURBO is a road-safety racing game. You steer a car through traffic on six different roads, collect coins to buy new cars, and every time you crash you learn a real road sign and the rule behind it.
Target Users	Children aged 8 to 14 who like racing games, teens who will drive soon, and teachers who want a fun start to a road-safety lesson.
Code	https://github.com/mr-basem/Turpo

How to Run | طريقة التشغيل

```
pip install pgzero
python Game.py     # from the project folder, so images/ and music/ are found
```

Controls | التحكم

Key or click	What it does
SPACE	Start a race from the Start screen
← / →	Steer left and right
P or Esc	Pause and resume
R	Race again after a game over
H	Go back to the Start screen from anywhere
F	Fullscreen on and off
🏠 / 🛒 / 👥 buttons	Home, car shop, game makers
🔊 button (on the pause screen)	Music on and off

Tech Stack | التقنيات المستخدمة

Category	Technology	Purpose
Language	Python 3	All of the game's logic
Game framework	Pygame Zero (`pgzero`)	Opens the 800 × 600 window, runs `draw()` and `update()` 60 times a second, loads images, music and sounds by name, and calls `on_key_down()` and `on_mouse_down()`
Graphics library	pygame (installed with Pygame Zero)	Rectangles for buttons and cards, fullscreen, the scaled window, and loading the makers' photos
Images	PNG and JPEG files in `images/`	6 roads, 6 cars, 8 shop cars, 20 road signs, icons and 4 team photos
Sound	MP3 and WAV files in `music/` and `sounds/`	Race music, game-over music and the coin "ding"

Dependencies | المكتبات

Package	Version	Purpose
pgzero	Not pinned (screenshots made with 1.2.1)	The game framework. There is no `requirements.txt` yet.
pygame	Comes with pgzero (2.6.1 here)	`Rect`, `toggle_fullscreen`, `image.load`, `transform.smoothscale`
random	Standard library	Picks lanes, car colours and the road sign

2. Problem Statement | المشكلة

Children walk, cycle and ride in cars on busy roads every day, long before they ever drive. Road signs are usually taught as a long list to memorise, so they are forgotten quickly. At the same time, the racing games children love reward speed and crashing, and teach nothing about driving safely.

Why does it matter? | لماذا هذا مهم؟

A sign you do not understand cannot protect you. Knowing what ”Stop”, ”Children crossing” or ”Slippery road” means helps a child cross the street, ride a bike and later drive more safely. Starting young builds habits that last.

How is it currently solved? | كيف تُحل المشكلة حاليًا؟

Today children learn road signs from school books, posters, or a driving school when they are much older. These are useful, but they are not fun, so children rarely go back to them. TURBO puts the lesson inside a game they already want to play again and again.

3. Technical Architecture | البنية التقنية

Project Structure | هيكل المشروع

```
Turpo/
├── Game.py              # Runs first: window, screen flags, draw(), update(), keys
├── start_page.py        # Start screen: logo, "Press SPACE", home and makers buttons
├── play_page.py         # The race: roads, levels, score, coins, traffic, hearts
├── pause_page.py        # Pause box and the music ON / OFF button
├── game_over_page.py    # The 20 road signs with their meanings, score and "Press R"
├── store.py             # The car shop: 9 cars, prices, owned and equipped car
├── makers_page.py       # "Game makers": the 4 team photos
├── images/
│   ├── roads/           # 6 road pictures, 800 × 800, one per level
│   ├── cars/            # The red player car and 5 traffic cars
│   ├── skins/           # 8 cars to buy in the shop
│   ├── signs/           # 20 road signs, 1.png to 20.png
│   ├── icons/           # Logo, coin, heart, home, shop, makers, volume and mute icons
│   └── makers/          # Team photos
├── music/               # game_music.mp3, game_over.mp3
├── sounds/              # coin_collect.wav (the coin "ding")
├── logo Game.png        # The full-size TURBO logo
├── team_logo.png        # The NINJA.PY team logo
└── docs/                # This document, the poster, the slides and the diagrams
```

Architecture and Data Flow | تدفق البيانات

```
python Game.py
    → pgzrun.go() opens an 800 × 600 window and starts the game loop
        → 60 times a second:
            update(dt) → if racing, play_page.update() moves everything
            draw()     → paints the road, then the chosen screen
        → a key:   on_key_down(key)   → SPACE, P, Esc, R, H, F
        → a click: on_mouse_down(pos) → home, shop, makers, volume, cars
```

Which screen shows is decided by a few true/false flags in `Game.py` (`game_started`, `paused`, `makers_open`, `store.is_open`) and by the hearts left (`play_page.health`). Each screen lives in its own file and has its own `draw()` function.

Key Files | الملفات الأساسية

File	Purpose
`Game.py`	Sets the window size, keeps the flags and the best score, and chooses which screen to draw. Starts and stops the music. Handles every key and click.
`play_page.py`	The biggest file (222 lines). Scrolls two copies of the road, fades to a new road at each level, moves the player, adds coins and traffic in 6 lanes, checks crashes, and draws the score, level, best score, hearts and coins.
`store.py`	The list of 9 cars with names and prices, the shop cards in a 3 × 3 grid, and buying or equipping a car when its card is clicked.

File	Purpose
`game_over_page.py`	The 20 road signs with a two-line meaning for each, and the game-over text.

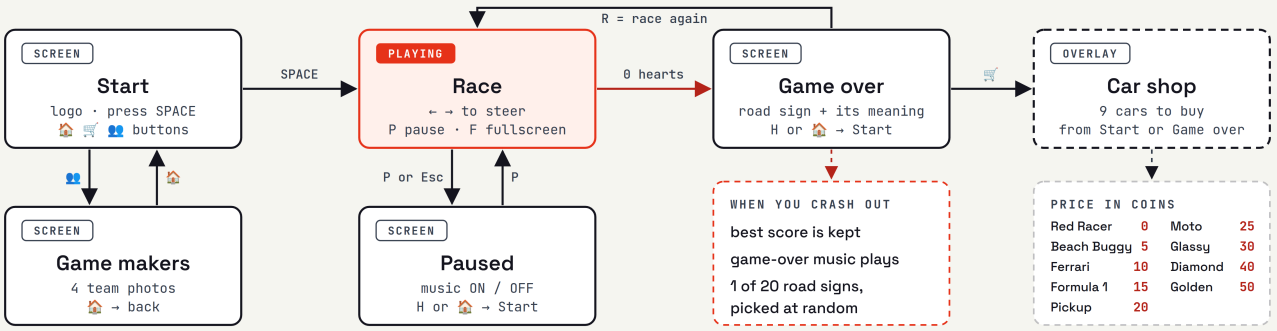
4. Features & User Flow | المميزات ورحلة المستخدم

Feature List | قائمة المميزات

#	Feature	Description	Status
1	Steering	← and → move the car 6 pixels a frame, only between the road's edges.	Complete
2	Traffic	Cars appear in one of 6 lanes, never two in the same lane close together. A crash costs 1 of 3 hearts, then the car blinks for 1 second and cannot be hit.	Complete
3	Score and levels	+1 point every half second. A new level every 30 points, up to level 6.	Complete
4	Six roads	Egypt, forest, snow, night, green fields and desert. The old road fades out over 0.75 seconds.	Complete
5	Getting harder	Every 10 points the speed goes up by 0.3 (top speed 15) and new cars come sooner (from every 2 seconds down to every 0.67 seconds).	Complete
6	Coins	A coin every 2.3 seconds. Touching it adds 1 coin and plays a "ding".	Complete
7	Car shop	9 cars, from the free Red Racer to the 50-coin Golden Car. Buy once, then equip any time.	Complete. Nothing is saved after the game closes (see Limits).
8	Road sign lesson	Every game over shows 1 of 20 real road signs, picked at random, with what it means and what to do.	Complete
9	Best score	The highest score is shown during the race as "Best".	Complete. Kept only while the game is open.
10	Pause and music	P or Esc pauses. The speaker button turns the music on or off.	Complete
11	Fullscreen	F switches fullscreen on and off. The window scales to the screen.	Complete
12	Game makers	A screen with the team's photos.	Complete

User Flow | رحلة المستخدم

SCREENS · CHOSEN BY TRUE/FALSE FLAGS IN GAME.PY AND THE HEARTS LEFT



EVERY FRAME OF A RACE · update() IN PLAY_PAGE.PY, 60 TIMES A SECOND



3 hearts at the start. When the last heart is gone, update() stops moving things and the Game over screen takes over.
Speeds are multiplied by the frame time, so the race runs the same on a slow or a fast computer.

```
Start (start_page.py)
    → SPACE → Race (play_page.py)
        → P or Esc → Paused (pause_page.py) → P → back to the race
        → 0 hearts → Game over (game_over_page.py): road sign + meaning
            → R → a new race (score and level reset, coins and cars kept)
            → H or 🏠 → Start
    → 🛒 → Car shop (store.py), also from Game over
    → 👥 → Game makers (makers_page.py) → 🏠 → Start
```

The Six Levels | المستويات الستة

SIX ROADS · A NEW ONE EVERY 30 POINTS (ABOUT 15 SECONDS)



LEVEL 1
Egypt

points	0–29
reached at	0:00
start speed	3.0
new car every	2.0 s



LEVEL 2
Forest

points	30–59
reached at	0:15
start speed	3.9
new car every	1.5 s



LEVEL 3
Snow

points	60–89
reached at	0:30
start speed	4.8
new car every	1.0 s



LEVEL 4
Night

points	90–119
reached at	0:45
start speed	5.7
new car every	0.67 s



LEVEL 5
Green fields

points	120–149
reached at	1:00
start speed	6.6
new car every	0.67 s



LEVEL 6
Desert

points	150+
reached at	1:15
start speed	7.5 → 15
new car every	0.67 s

Faster every 10 points
The road, coins and cars speed up by 0.3. Top speed is 15, at 400 points.

More traffic
A new car every 2 seconds at the start. From 80 points: every 0.67 seconds, in one of 6 lanes.

Smooth road change
The old road fades out over 0.75 seconds while the new one scrolls in.

Level	Road	Points	Reached after	Speed at the start	New car every
1	Egypt	0–29	0:00	3.0	2.0 s
2	Forest	30–59	0:15	3.9	1.5 s

Level	Road	Points	Reached after	Speed at the start	New car every
3	Snow	60–89	0:30	4.8	1.0 s
4	Night	90–119	0:45	5.7	0.67 s
5	Green fields	120–149	1:00	6.6	0.67 s
6	Desert	150 and up	1:15	7.5, rising to 15 at 400 points	0.67 s

Times assume 60 frames a second. The game multiplies every move by the frame time, so it runs at the same speed on slow and fast computers.

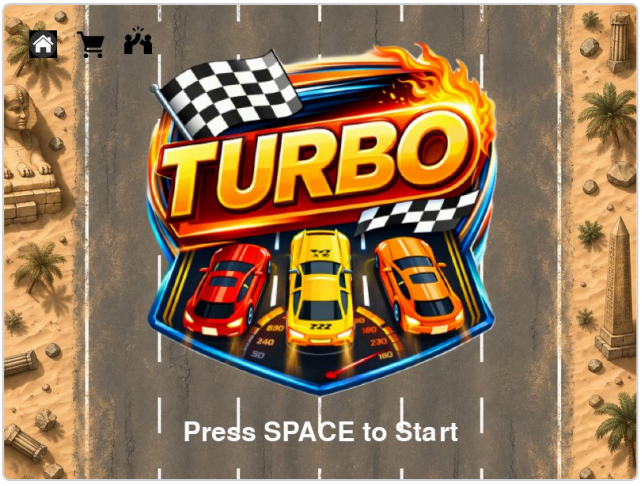
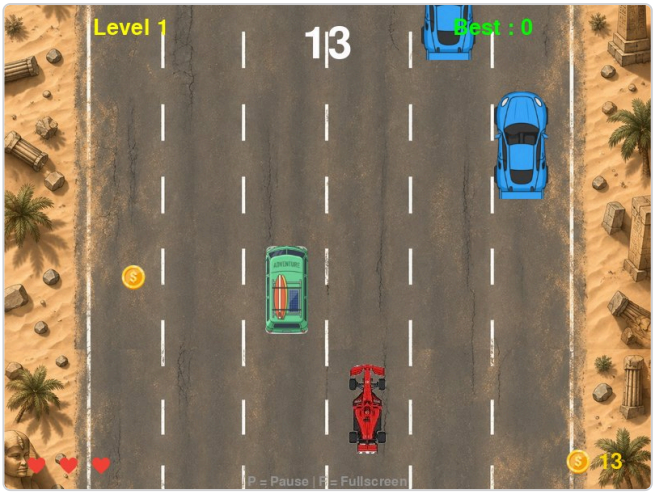
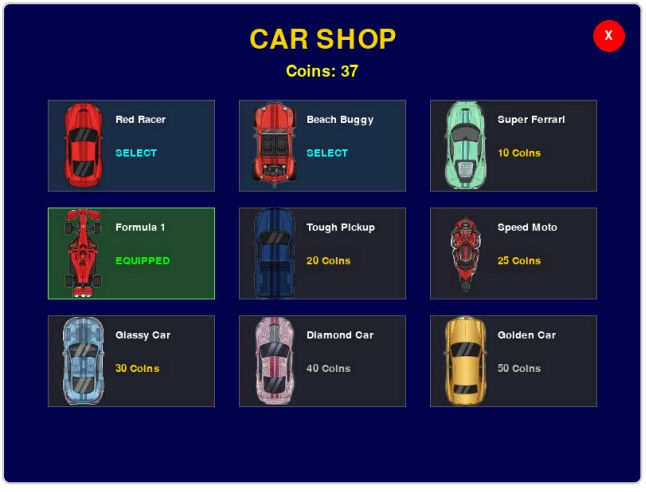
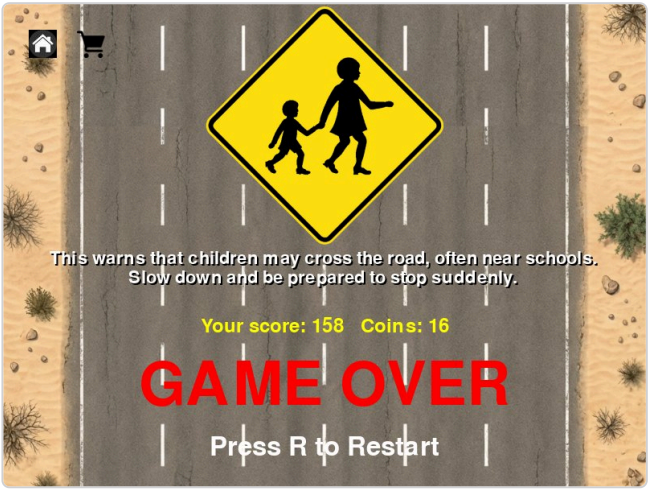
Screen Reference | الشاشات

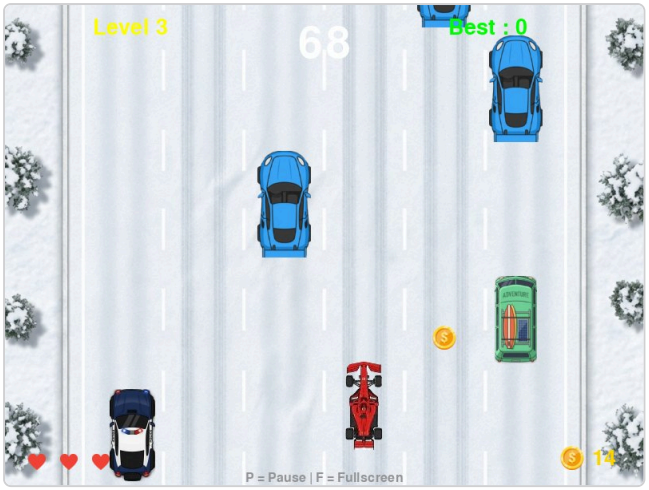
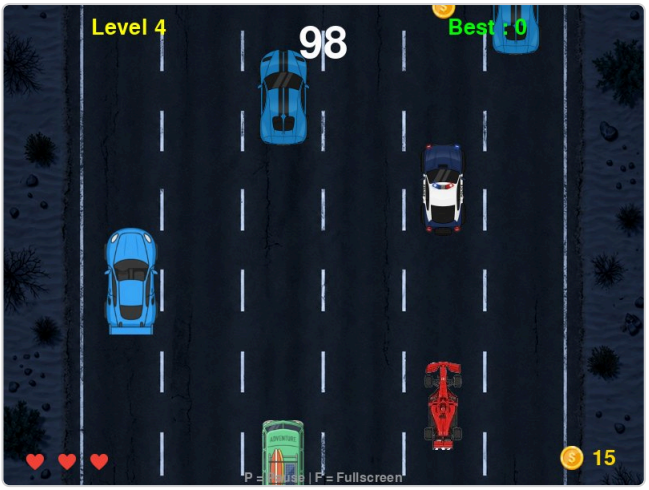
Screen	File	Description
Start	`start_page.py`	The TURBO logo over the Egypt road, "Press SPACE to Start", and the home, shop and makers buttons.
Race	`play_page.py`	The road, your car, traffic and coins. Score in the middle, level and best score on top, hearts and coins at the bottom.
Paused	`pause_page.py`	A black box with "PAUSED" and "Press P to Resume", the music button, and the home button.
Game over	`game_over_page.py`	A road sign and its meaning, your score and coins, "GAME OVER" and "Press R to Restart".
Car shop	`store.py`	9 cards with each car, its name, and its price, "SELECT" or "EQUIPPED". A red X closes it.
Game makers	`makers_page.py`	The team's photos in a 2 × 2 grid and "Thanks for playing!".

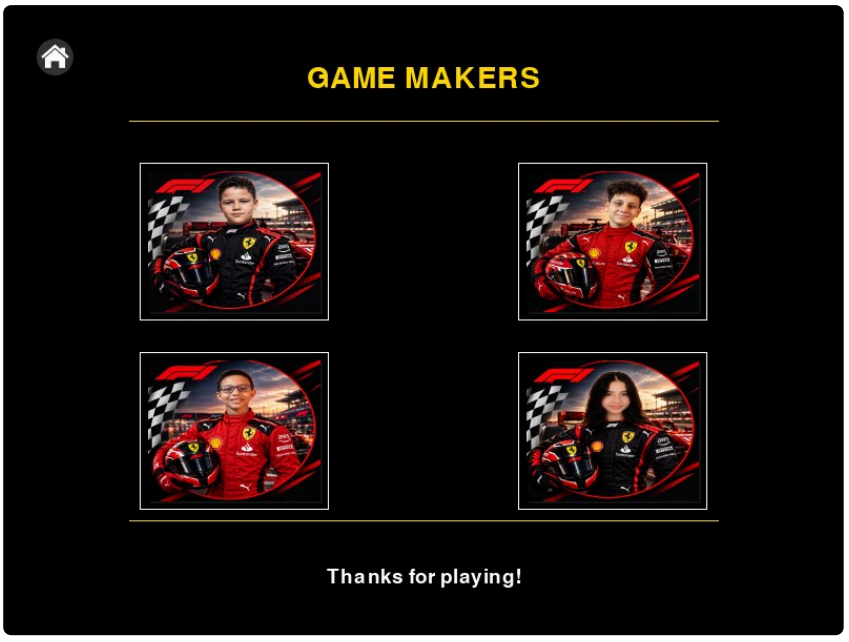
5. UI/UX Design | تصميم واجهة المستخدم

Screenshots | لقطات الشاشة

These are real screenshots of the game, captured from its own `draw()` function.

Start	Race (level 1, Egypt)
	
Race (level 2, forest)	**Paused**
	
Car shop	**Game over**
	

Snow (level 3)	Night (level 4)
	



Get the code: https://github.com/mr-basem/Turpo

Design System | نظام التصميم

Element	Current implementation
Logo	"TURBO" in big yellow-orange letters with flames, checkered flags, a speedometer and three racing cars (red, yellow, orange) on a road (`images/icons/logo_game.png`, full size in `logo Game.png`).

Element	Current implementation
Window	800 × 600 pixels, scaled to fit the screen (`pygame.SCALED`), with fullscreen on F.
View	Top-down: the road scrolls down and the player's car stays near the bottom. Traffic cars are turned 180°.
Score	Big white number at the top centre.
Level / Best	"Level N" in yellow at the top left, "Best : N" in lime green at the top right.
Hearts and coins	Red hearts at the bottom left, a gold coin and the coin count at the bottom right.
Pause box	Black box with a yellow border, "PAUSED" in yellow.
Car shop	Dark blue background `#00034D`, "CAR SHOP" in gold. Equipped car: green card. Owned car: "SELECT" in cyan. Prices in gold when you can afford them, grey when you cannot.
Game over	"GAME OVER" in red, score and coins in yellow, the sign's meaning in white with a black shadow.
Font	Pygame Zero's built-in font. The poster and slides use Russo One, Space Grotesk and JetBrains Mono.

6. Data Sources | مصادر البيانات

Data Sources | مصادر البيانات

TURBO has no database and no internet connection. All of its data is in the code and in the `images/` and sound folders.

Source	Type	Description
`SKINS` in `store.py`	List in the code	The 9 cars: picture, name and price.
`ROAD_SIGNS` in `game_over_page.py`	Dictionary in the code	The 20 road signs: picture number and a two-line meaning.
`BACKGROUNDS` in `play_page.py`	Dictionary in the code	Which road picture belongs to each of the 6 levels.
`images/`	Picture files	Roads, cars, signs, icons and team photos.
`music/`, `sounds/`	Sound files	Race music, game-over music and the coin sound.
The player	Keys and clicks	Steering, pausing, buying and equipping cars.

Game Data | بيانات اللعبة

The car shop (`store.py`)

Car	Picture	Price
Red Racer	`cars/car_red`	Free
Beach Buggy	`skins/beachbugyy`	5
Super Ferrari	`skins/ferrari`	10
Formula 1	`skins/formula1`	15
Tough Pickup	`skins/pickup`	20
Speed Moto	`skins/motocycle`	25
Glassy Car	`skins/glassycar`	30
Diamond Car	`skins/diamound`	40
Golden Car	`skins/golden_car`	50

The 20 road signs (`game_over_page.py`)

#	Sign	#	Sign
1	Road works	11	No left turn
2	Stop	12	Steep hill
3	Slippery road	13	Workers on the road
4	Falling rocks	14	Roundabout ahead
5	Pedestrian crossing	15	Uneven road
6	No parking	16	Give way to oncoming traffic
7	No entry	17	Children crossing
8	Sharp curves	18	Cyclists
9	Speed hump	19	Reduce speed
10	No overtaking	20	Traffic lights ahead

What the game remembers while it is open

Variable	File	What it holds
`high_score`	`Game.py`	The best score so far
`coins`	`play_page.py`	Coins collected. Kept when you race again.
`owned_skins`, `selected_skin`	`store.py`	The cars you bought and the one you drive
`score`, `level`, `health`, `speed`	`play_page.py`	The current race. Reset when you race again.

External APIs | الخدمات الخارجية

API	Purpose	Rate Limit
None	The game runs fully offline.	Not applicable

Limits | حدود المشروع

These shortcuts keep the code short and simple:

- Nothing is saved to a file. Coins, cars you bought and the best score are gone when the game closes.
- There is no `requirements.txt`, so the Pygame Zero version is not fixed.
- Run `python Game.py` from the project folder. The makers' photos are loaded with a path that starts from the folder you are in.
- The makers screen loads its 4 photos from the disk again on every frame.

Known Gaps | ملاحظات للتحسين

Found while writing this document:

- During a race, the home and shop buttons are not drawn, but clicking where they would be still works. A click near the top left can open the car shop while the race keeps running underneath it.
- On the snow road (level 3), the yellow "Level 3" text is hard to read on the white snow.
- Two files are not used by the code: `images/icons/game_over_logo.png` and `sounds/coin_collect.mp3` (Pygame Zero plays `coin_collect.wav`).

7. Educational Content | المحتوى التعليمي

Learning Objectives | أهداف التعلم

Through building this project, a student can practice:

1. How a game loop works: `update()` changes the game and `draw()` paints it, 60 times a second.
2. Using x and y coordinates to place and move pictures, and checking when two pictures touch (crashes and coins).
3. Using true/false flags to decide which screen to show: start, race, pause, game over, shop and makers.
4. Making a game fair and fun by tuning numbers: speed, how often cars come, prices and level length.
5. Road safety: what 20 common road signs mean and what a driver or walker should do when they see them.

Cultural / Historical Context | السياق الثقافي

Topic	Description
Road safety in Egypt	Egypt's roads are busy, and many people walk next to traffic. Knowing road signs from a young age helps children stay safe as walkers, cyclists and future drivers.
Egyptian landscapes	The first road is an Egyptian road with sand, palm trees and old stone monuments, and the last one is the desert, so players race through places that look like home.
Racing and Formula 1	Formula 1 is loved all over the world. The shop's Formula 1 car and the team's racing-suit photos bring that excitement into a game about safe driving.

Curriculum Alignment | الارتباط بالمنهج

Skill Area	What the Student Practices
Problem Solving	Finding and fixing bugs, such as the music not starting again after a game over.
Computational Thinking	Breaking the game into small files and steps: move the road, move the car, add cars, check crashes.
Creativity	Designing six different roads, nine cars and the TURBO logo.
Collaboration	Jomana and Ali shared the work as Team NINJA.PY: screens, pictures, road signs and testing.
Technical Writing	Writing a clear, short meaning for each of the 20 road signs.

8. Marketing Plan | خطة تسويقية

Target Audience | الجمهور المستهدف

- **Primary:** Children aged 8 to 14 who enjoy racing games.
- **Secondary:** Teachers, schools and parents who want a fun way to teach road safety.

Platforms | المنصات

- Windows, macOS and Linux computers with Python installed. The code is free on GitHub.
- School computer labs, road-safety days, and tech fairs such as the Afro-Asian Forum for Innovation and Technology (AFFIAT).

9. Student Worksheet | ورقة عمل الطالب

My Project Information | معلومات مشروعي

Student Names: Jomana and Ali (Team NINJA.PY)
Project Title: TURBO: Top-Down Racing Game

What is your project about? | عن ماذا يتحدث مشروعك؟

TURBO is a racing game where you dodge cars on six roads and collect coins to buy new cars. When you crash, the game teaches you a real road sign.

What problem does it solve? | ما المشكلة التي يحلها؟

Children find road signs boring to learn from a book. TURBO teaches them inside a game they want to play again and again.

My Data | بياناتي

Where does data come from?

We made the lists ourselves in the code: the 9 cars with their prices, the 20 road signs with their meanings, and the 6 roads. The pictures and sounds are files in the project folders.

What data is stored?

Nothing is saved to a file. While the game is open it remembers your coins, the cars you bought, the car you drive and your best score.

What I Learned | ماذا تعلمت

1. **The hardest part:** Making the road scroll forever and change smoothly to a new road at each level.
2. **The most fun part:** Designing the car shop and watching the game get faster and harder.
3. **If I had more time:** Save coins, cars and the best score to a file, and add more road signs and roads.
4. **One thing I would do differently:** Plan all the screens and buttons first, so hidden buttons cannot be clicked during a race.

Screenshot of My Project | لقطة من مشروعي